

**(a) measurements deviate from the norm:
measurements decide (video was wrong)**



trunk lean (°)	34.20	z +3.6
shoulder offset	0.54	z +3.8
head forward (°)	59.40	z +2.1
hip flexion max (°)	83.56	z +3.8
hip range (°)	46.95	z +1.7

mean $|z| = 3.18 \geq \text{threshold } 0.52$
 → trust measurements (w = 0.75)
 P(ASD) video 0.40 meas. 0.99 fused 0.84
 truth ASD gate ✓ video ✗ meas. ✓

**(b) measurements unremarkable:
video decides (measurements were wrong)**



trunk lean (°)	2.31	z -0.1
shoulder offset	0.04	z -0.2
head forward (°)	18.06	z -0.3
hip flexion max (°)	36.28	z -0.3
hip range (°)	30.99	z +0.0

mean $|z| = 0.35 < \text{threshold } 0.53$
 → trust video (w = 0.00)
 P(ASD) video 0.97 meas. 0.47 fused 0.97
 truth ASD gate ✓ video ✓ meas. ✗

(c) failure case



trunk lean (°)	14.36	z +1.3
shoulder offset	0.25	z +1.5
head forward (°)	23.42	z -0.0
hip flexion max (°)	52.65	z +1.1
hip range (°)	36.28	z +0.6

mean $|z| = 0.92 \geq \text{threshold } 0.53$
 → trust measurements (w = 1.00)
 P(ASD) video 0.48 meas. 0.94 fused 0.94
 truth non-ASD gate ✗ video ✓ meas. ✗